UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

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Course Name : ____________________

Lecturer : ____________________

Academic Session : ____________________

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1 Template Overview

This document is a demo for the **UIUC Assignment Template** — a general-purpose LaTeX template for English assignments at University of Illinois Urbana-Champaign (UIUC).

Features provided by `uiuc_asg.cls`:

- UIUC-style header and "Page X of Y" footer
- APA citation style (`\citet{}` / `\citep{}`) via `apacite`
- Syntax-highlighted code blocks
- Smart cross-references with `\cref{}` / `\Cref{}` (`cleveref`)
- Figure and table numbering per section (Figure 1.1, Table 2.3, …)
- Cover page / Rubric / Turnitin report attachment via `\includepdf`
- `[turnitin]` class option — suppress headers/footers for Turnitin submission

1.1 Recommended Project Structure

Organise your assignment folder as shown below. This layout separates source content, images, and university documents into clearly defined locations.

```
your-assignment/
├── main.tex                <- Entry point; edit metadata, then \input sections
├── uiuc_asg.cls            <- Template class; do not edit unless necessary
├── ref.bib                 <- BibTeX references (APA format)
│
├── sections/               <- One .tex file per section (strongly recommended)
│   ├── 01_introduction.tex
│   ├── 02_methodology.tex
│   └── ...
│
├── figure/                 <- All images go here
│   └── my_chart.png
│
└── docs/                   <- Official documents from the university
    ├── cover_page.pdf      <- Exported from cover_page.docx
    ├── sample_rubric.pdf   <- Marking rubric from the course portal
    └── turnitin_report.pdf <- Downloaded after Turnitin submission
```

1.2 Modular Section Management

Strongly recommended: split your content into one `.tex` file per section and `\input` them from `main.tex`:

```
\input{sections/01_introduction.tex}
\input{sections/02_methodology.tex}
\input{sections/03_results.tex}
```

Do *not* write all your content directly inside `main.tex`.

Why? This approach makes AI-assisted writing significantly more effective. AI tools such as ChatGPT, Claude, and Gemini operate within a limited context window. When your entire document is one long file, the AI can only see a portion of it at a time, often missing critical context from earlier or later sections. By feeding one section file at a time, the AI reads your complete section in full, produces higher-quality suggestions, and edits more accurately without wasting its context budget on unrelated content.

1.3 Compilation

This template requires **XeLaTeX** (not pdfLaTeX). The number of passes depends on whether your document contains citations.

With citations (i.e. you use `\bibliography{ref}`):

```
xelatex main   (1st pass — generates .aux)
bibtex main    (resolves citations from ref.bib)
xelatex main   (2nd pass — inserts bibliography)
xelatex main   (3rd pass — resolves all cross-references)
```

Without citations (no `\bibliography{}` in your document):

```
xelatex main   (1st pass)
xelatex main   (2nd pass — resolves cross-references and ToC)
```

In **VS Code** with the LaTeX Workshop extension, set the recipe to `latexmk` — it detects citations automatically and runs the correct number of passes without manual intervention.

2 Figures

Figures are inserted with the `figure` environment. The `[H]` specifier (from the `float` package) forces the figure to appear *exactly here* in the document, rather than floating to the top or bottom of the page.

Figure 2.1: *Official crest of University of Illinois Urbana-Champaign (UIUC).*

Cross-reference example: The university crest is shown in **??**. Use `\Cref{}` (capital C) at the start of a sentence and `\cref{}` (lowercase c) in the middle of one.

3 Tables

A professional three-line table uses `booktabs` rules: `\toprule`, `\midrule`, and `\bottomrule`. Avoid using vertical lines — they are considered poor typographic practice.

Table 3.1: *Comparison of common programming paradigms.*

Paradigm	**Key Concept**	**Example Language**	**Typical Use Case**
Procedural	Step-by-step execution	C, Pascal	System-level programming
Object-Oriented	Encapsulation, classes	Java, Python	Application development
Functional	Immutability, pure functions	Haskell, Clojure	Data transformation
Declarative	What, not how	SQL, Prolog	Querying, logic reasoning

As shown in Table 3.1, different paradigms suit different problems.

4 Citations

This template uses the **APA** citation style via `apacite`. Two citation commands are available:

- `\citet{}` — *narrative* citation, where the author's name is part of the sentence.
- `\citep{}` — *parenthetical* citation, placed in brackets at the end of a clause.

Narrative citation example:

? introduced the Transformer architecture, demonstrating that self-attention mechanisms alone are sufficient for state-of-the-art sequence modelling without recurrence or convolution.

Parenthetical citation example:

The Transformer model has since become the dominant architecture for natural language processing tasks, underpinning large language models such as GPT and BERT (?).

5 Code Listings

The template provides syntax-highlighted environments for common languages.

5.1 Python

```python
# A simple example using the Transformer attention concept
import torch
import torch.nn.functional as F

def scaled_dot_product_attention(Q, K, V):
    """Compute scaled dot-product attention."""
    d_k = Q.size(-1)
    scores = torch.matmul(Q, K.transpose(-2, -1)) / (d_k ** 0.5)
    weights = F.softmax(scores, dim=-1)
    return torch.matmul(weights, V)
```

5.2 SQL

```sql
-- Retrieve top-5 students by GPA
SELECT student_id, name, gpa
FROM   students
WHERE  status = 'active'
ORDER  BY gpa DESC
LIMIT  5;
```

6 Best Practices

The tips in this section come from a year of using this template across UIUC assignments of all sizes — from 10-page weekly reports to a 500-page community service final report. Following these practices will save you time and help you avoid common pitfalls.

6.1 Modular File Structure (Strongly Recommended)

Rather than writing everything inside a single `main.tex`, split your content into one `.tex` file per section and use `\input{}` to include them:

```
main.tex
sections/
    01_introduction.tex
```

```
02_methodology.tex
03_results.tex
04_discussion.tex
05_conclusion.tex
```

This approach has several advantages:

- **AI-assisted writing is more effective.** AI tools (ChatGPT, Claude, Gemini, etc.) have limited context windows. Feeding one section at a time means the AI can read your entire section without truncation, resulting in higher-quality suggestions and edits.
- **Easier collaboration.** Teammates can work on different section files simultaneously without merge conflicts.
- **Faster debugging.** When a compilation error occurs, the error message will point to the specific section file rather than a line number in a 1000-line `main.tex`.

6.2 Fast Compilation for Image-Heavy Documents

If your document contains many figures, compilation can become slow because XeLaTeX re-processes every image on every run.

Solution: Use the `draft` class option during writing. In draft mode, figures are replaced by placeholder boxes, reducing compile time from minutes to seconds:

```
\documentclass[draft]{uiuc_asg}   % Fast: figures shown as boxes
\documentclass{uiuc_asg}          % Normal: figures rendered fully
```

Switch back to the normal option only for your final compilation.

6.3 Compiling for Turnitin Submission

Problem: Turnitin scans every page of your PDF, including headers and footers. Because this template places `UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN` in the header and `Page X of Y` in the footer of *every* page, Turnitin may flag these repeated strings as similarity matches, artificially inflating your similarity score.

Solution: This template provides a dedicated `[turnitin]` class option that disables all headers and footers in one step. When you are ready to generate the PDF for Turnitin submission, change the very first line of `main.tex`:

```
% Normal version (for your own records and lecturer submission):
\documentclass{xum_review}
```

```
% Turnitin version (for upload to the originality checker):
\documentclass[turnitin]{xum_review}
```

Compile the Turnitin version, upload the resulting PDF, then switch back to the normal version for your final lecturer submission.

- This is a **one-word change** — no need to comment out multiple lines of header/-footer configuration.
- The body content, margins, and all formatting remain identical between the two versions.
- Your lecturer's copy (with the UIUC header) and your Turnitin copy are compiled from the same source — no risk of accidentally submitting different versions.

A Appendix Sample

Sample appendix.